



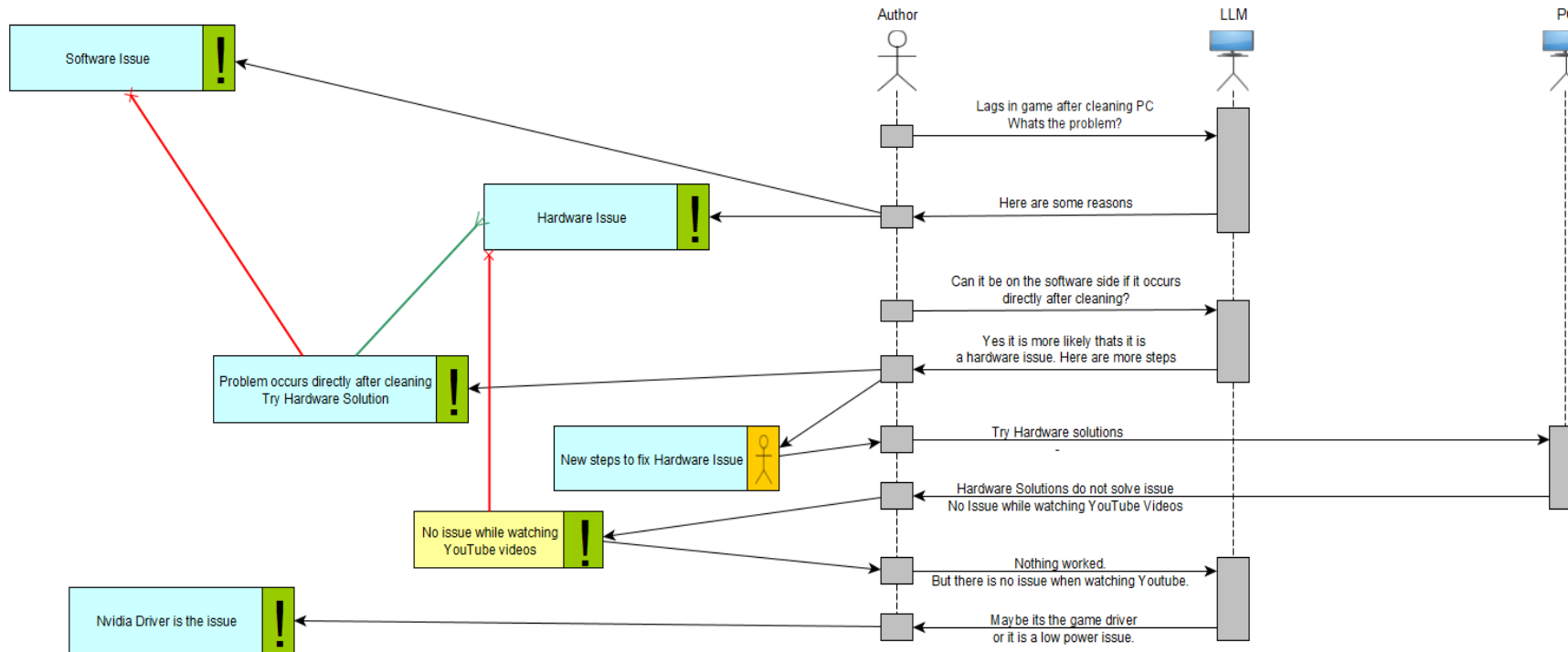
➤ KEML Improvement: Recursive Edges

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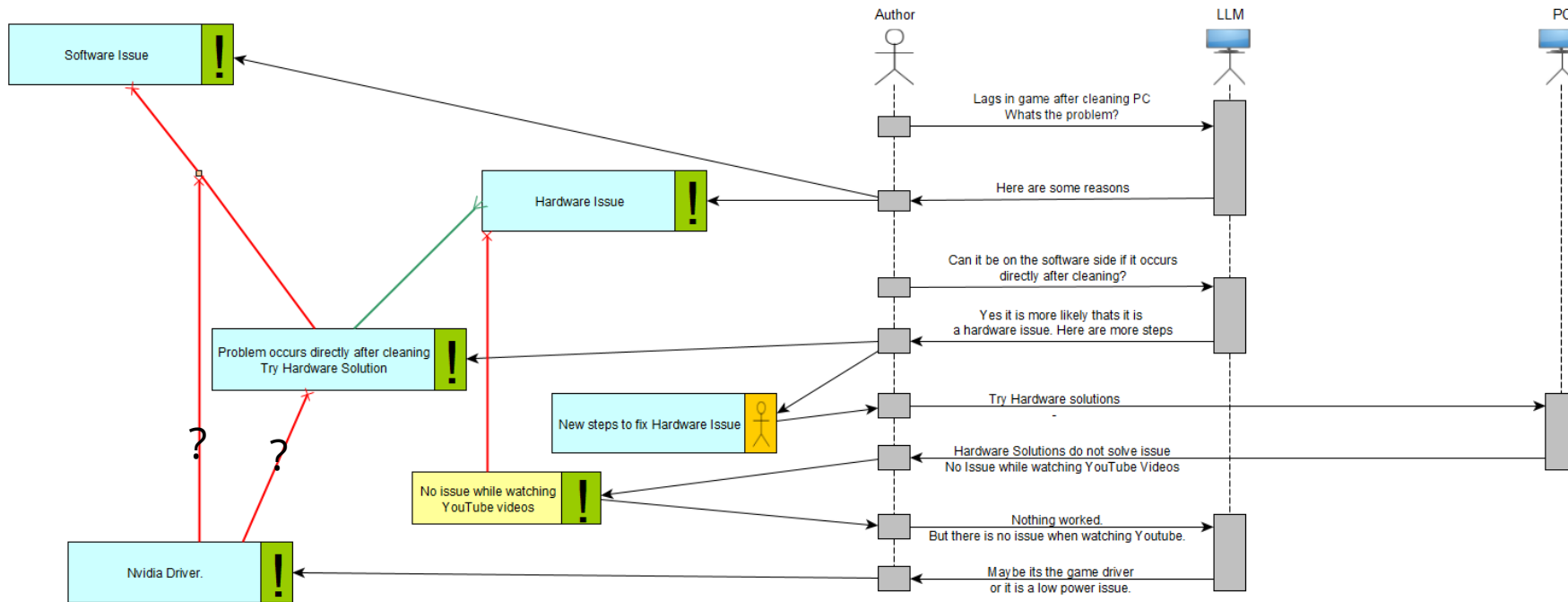
Motivation



- Task for ChatGPT:
- Find the issue that causes lag when playing games on a PC
 - Is it a hardware or software issue?

→ What should
"Nvidia Driver is the
Issue" attack?

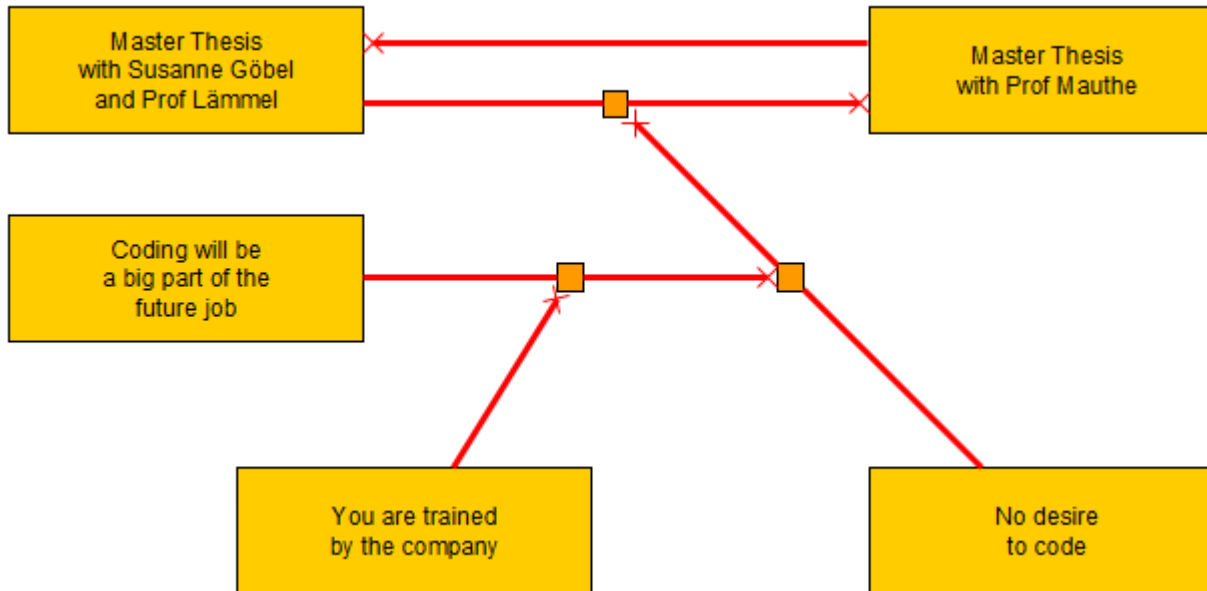
Motivation



1. Attack Argument
 - Stronger impact on "Software Issue"
 - Does it really attack "Problem occurs after cleaning"?
2. Attack Edge
 - Weaker impact on "Software Issue"

→ We will see the trust change later in the trust scores

How Recursive Edges are used in Literature



- Modeling decision making
- Recursive attacks cancel attacks out

→ Limitations for KEML

- No provable solution
- No weights

How Recursive Edges are used in Literature

“AFRA: Argumentation framework with recursive attacks” (Baroni et al. 2010)

- Introduction to recursive attacks
- Uses AFRA (Argumentation Framework with Recursive Attacks) instead of BAF
- Used for decision making

“Characterizing acceptability semantics of argumentation frameworks with recursive attack and support relations” (Gottifredi et al. 2016)

- Introduction to recursive attacks **and** supports
- Uses ASAF (Attack-Support Argumentation Framework)
- Used for decision making

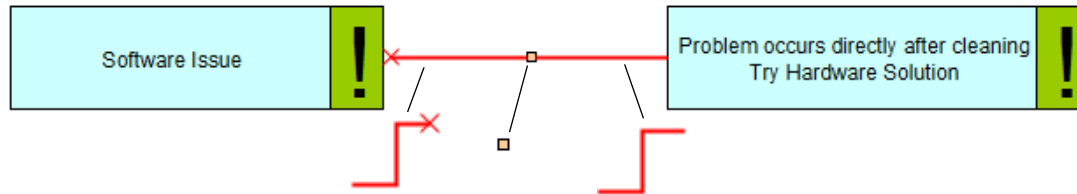
“Dynamics in Abstract Argumentation Frameworks with Recursive Attack and Support Relations” (Alfano et al. 2020)

- Update an existing model
- Uses ASAF

→ The frameworks do not contain weights and only used for decision making

→ Recursive attacks makes links invalid, recursive supports can be necessary for links to hold

- Creating new edges (recursive links) in yEd

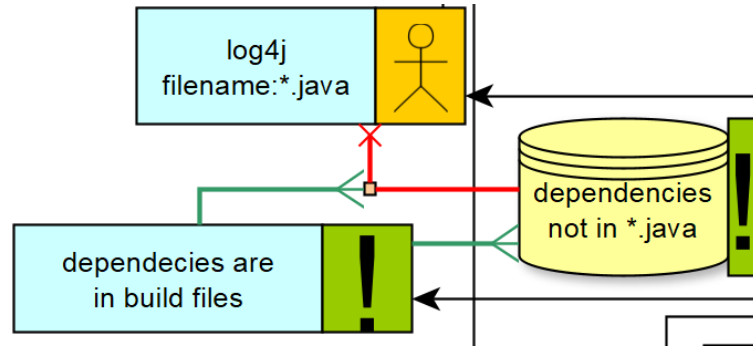


-> This is done by diving edge with **intermediate node**

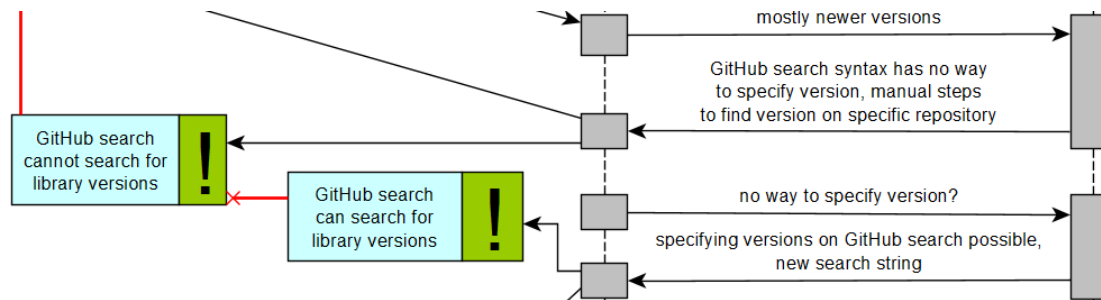
- Deciding whether an edge attacks/supports an information or an edge
 - Change the trust of a link or an information?
- More modelling choices for “find solution of problem” -task

Running Example 3.2

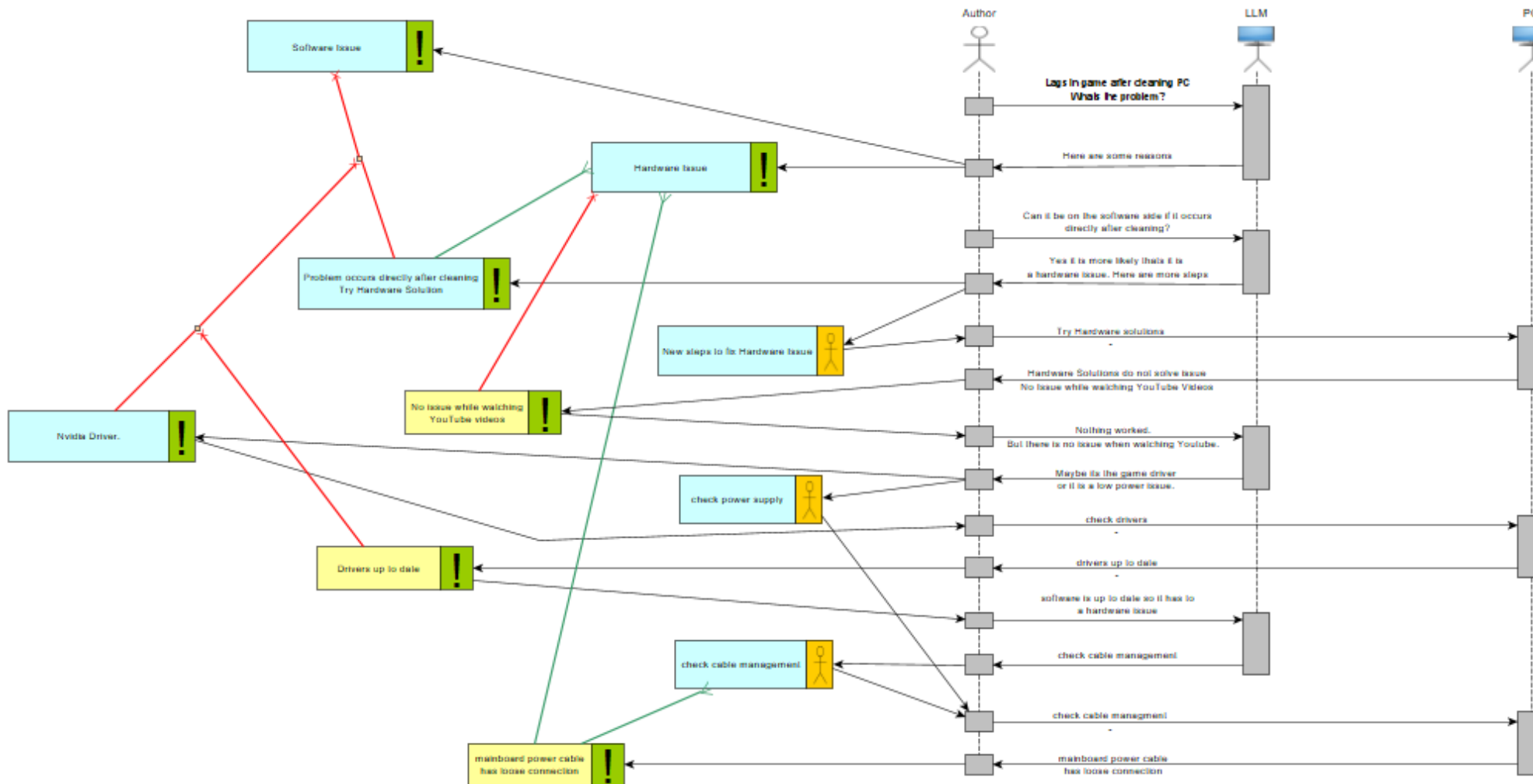
- Works with new framework and some recursive nodes can be added
 - Example: Strengthen the effect of the attack



- But 3.2 is not suitable for the framework
 - Example: In this case an attack on an attack is more suitable



Find PC Issue Example



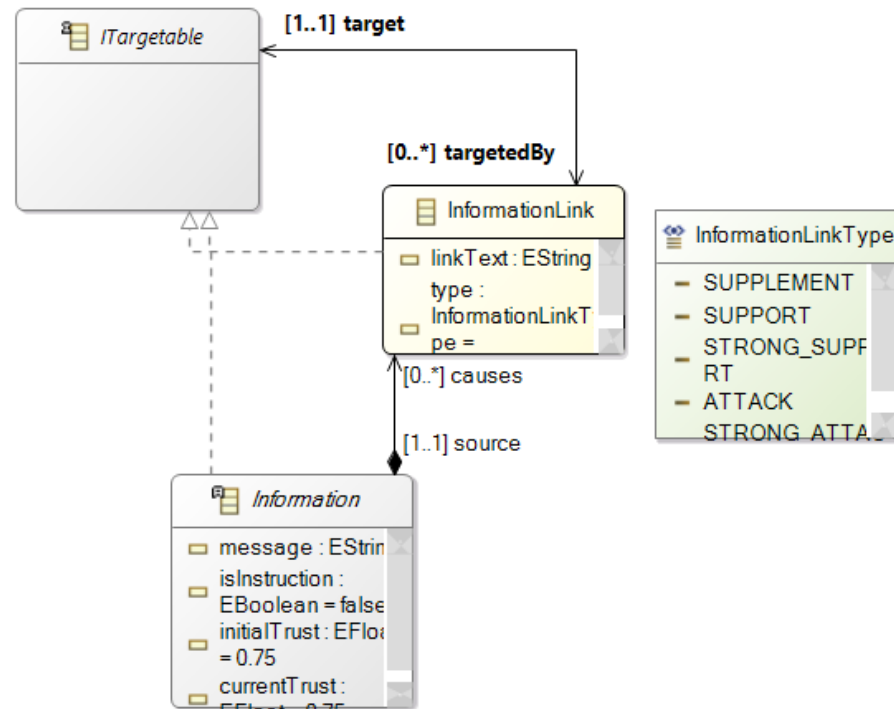
Find PC Issue Example

- ChatGPT should find a solution for the task: “Why does the PC lag?”
- It narrow it down to two possible issues
- Gives instruction to narrow down the possible causes
- Finds the issue (loose connection)

- 
- Recursive edges helps to model the dynamic of finding a solution
 - Trust is calculated for the whole conversation

Implementation – Changes for keml, keml.io and keml.analysis

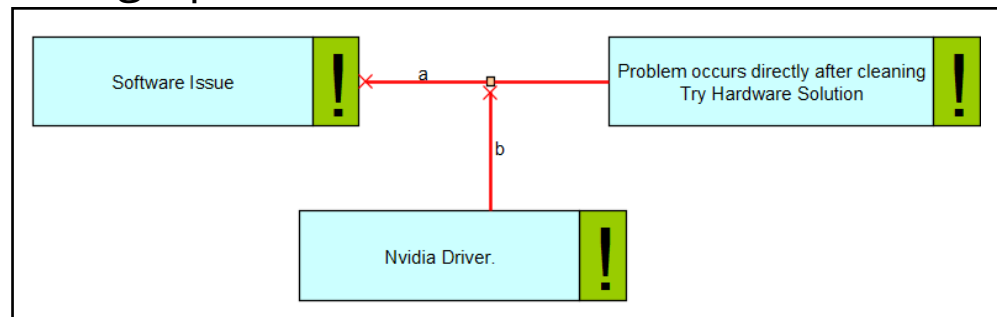
- Metamodel (keml)
 - Added interface `ITargetable`
→ Target of an `InformationLink` can be `Information` or `InformationLink`



Implementation – Changes for keml, keml.io and keml.analysis

- keml.io
 - Connect edges that are divided by intermediate node
 - Connect recursive edges with their target edge

/graphml/



keml.io

/keml/

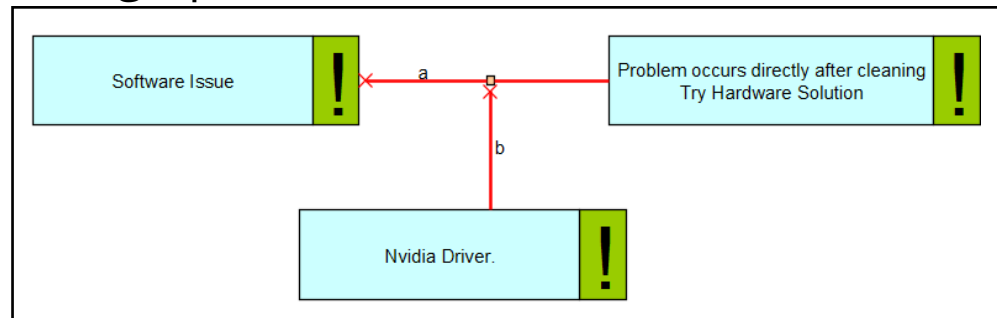
a	
Property	Value
Link Text	a
Source	New Information Problem occurs directly after cleaning Try Hardware Solution
Target	New Information Software Issue
Targeted By	Information Link b
Type	STRONG_ATTACK

b	
Property	Value
Link Text	b
Source	New Information Nvidia Driver.
Target	Information Link a
Targeted By	Information Link
Type	STRONG_ATTACK

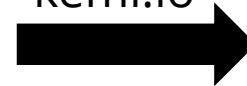
Implementation – Changes for keml, keml.io and keml.analysis ***NEW***

- keml.io
 - Connect edges that are divided by intermediate node
 - Connect recursive edges with their target edge

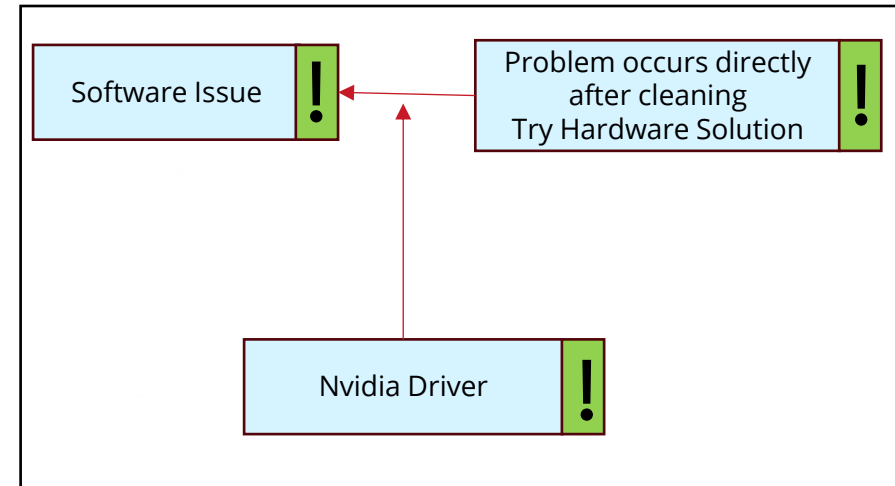
/graphml/



keml.io



/kempl/



Implementation – Changes for keml, keml.io and keml.analysis ***NEW***

Connect splitted edge

- Source of edge originating from intermediate node (e2) is set to the source of the edge that goes into the intermediate node (e1)
- Edge e2 is transformed to KEML edge

Create recursive Edge in KEML

- Relation of two edges is saved in list (e.g. r1 -> e2)
- All non-recursive edges that are already created are checked whether they are the target of a recursive edge
 - If this is true, the recursive link is created with the already created KEML edge as target

Implementation – Changes for keml, keml.io and keml.analysis

- keml.analysis
 - Added row to *arguments.csv* that counts recursive edges

KnowledgePart			
Information	LLM	PC	Author
Facts		4	3
Instructions		3	0
Repetitions		0	
RecAttacks		2	

- Added entries for number of recursive edges between conversation partner for *general.csv*

Attacks/Supports/RecAttacks/RecSupports	LLM F	LLM I	PC F	PC I	Author F	Author I
LLM F	1/1/1/0	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0
LLM I	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0
PC F	1/1/1/0	0/1/0/0	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0
PC I	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0
Author F	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0
Author I	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0

- Adds factor to trust analysis in *wX—trust.csv*

Trust Analysis – Strengthen and weaken Effects

Recursive Attacks

- Decrease trust score of a link
 - STRONG_ATTACK: (effect of targeted link)* $\frac{1}{2}$
 - ATTACK: (effect of targeted link)* $\frac{3}{4}$
 - Special case attack on recursive attack:
 - reduce the weakening of the recursive attack

Recursive Supports

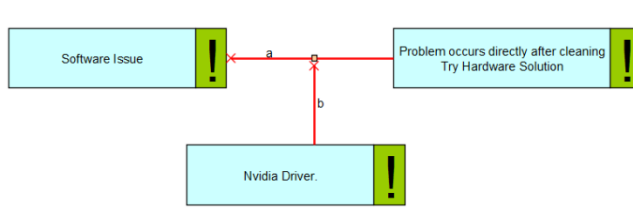
- Increase trust score of a link
 - STRONG_SUPPORT: (effect of targeted link)*2
 - SUPPORT: (effect of targeted link)* $\frac{4}{3}$

Trust Calculation

- NodeTrust = limitTo1(InitialTrust + RepetitionScore + w*argumentationScore)
- argumentationScore = $\Sigma(\text{edgeWeight} * \text{NodeTrust}(\text{source}) * \text{recEdgeFactor})$

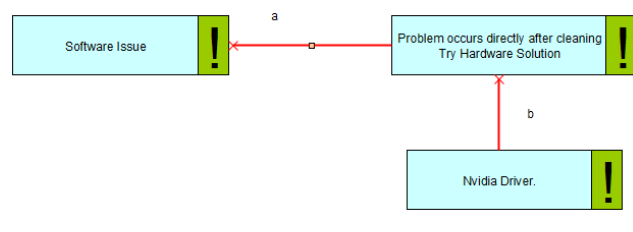
Trust Analysis – Attacking an Information vs. attacking an edge

Attack on edge



Time	Message	#Arg	#Rep	fTi	fTa	a		b		c		d	
						iT	T	iT	T	iT	T	iT	T
1	Software Issue	1	0			1.00	0.00	0.50	0.00	1.00	0.00	0.50	0.00
3	Problem occurs directly after cleaning Try Hardware Solu	0	0			1.00	1.00	0.50	0.50	1.00	1.00	0.50	0.50
7	Nvidia Driver.	0	0			1.00	1.00	0.50	0.50	1.00	1.00	0.50	0.50

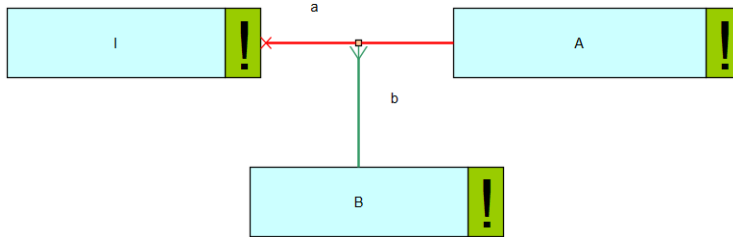
Attack on information



Time	Message	#Arg	#Rep	fTi	fTa	a		b		c		d	
						iT	T	iT	T	iT	T	iT	T
1	Software Issue	1	0			1.00	1.00	0.50	1.00	1.00	1.00	0.50	1.00
3	Problem occurs directly after cleaning Try Hardware Solu	1	0			1.00	-1.00	0.50	-0.50	1.00	-1.00	0.50	-0.50
7	Nvidia Driver.	0	0			1.00	1.00	0.50	0.50	1.00	1.00	0.50	0.50

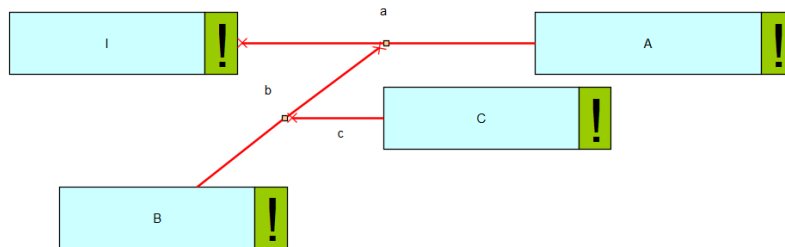
Trust Analysis – Additional Cases

Supporting an attacking edge



Time	M	#Arg	#Rep	fTi	fTa	a		b		c		d	
						iT	T	iT	T	iT	T	iT	T
1	A	0	0			1.00	1.00	0.50	0.50	1.00	1.00	0.50	0.50
1	I	1	0			1.00	-1.00	0.50	-1.00	1.00	-1.00	0.50	-1.00
3	B	0	0			1.00	1.00	0.50	0.50	1.00	1.00	0.50	0.50

Attacking a recursive attack



Time	M	#Arg	#Rep	fTi	fTa	a		b		c		d	
						iT	T	iT	T	iT	T	iT	T
1	A	0	0			1.00	1.00	0.50	0.50	1.00	1.00	0.50	0.50
1	I	1	0			1.00	-0.50	0.50	-0.25	1.00	-0.50	0.50	-0.25
3	B	0	0			1.00	1.00	0.50	0.50	1.00	1.00	0.50	0.50
5	C	0	0			1.00	1.00	0.50	0.50	1.00	1.00	0.50	0.50

Benefits

- New way of modelling the effect of trust
 - on information
 - on edges
- Better modeling approach for “find solution of problem”-tasks
- Better comparability (due to confidence values other than 0, |0.5| and |1.0|)
- Attacks on attacks are intuitive (also for non-computer scientist)

Limitations

- Few areas of application
- Trust calculation differs from literature
- Inconvenient modelling in yEd
 - Maybe KEML web-editor can improve this

Icons

Camel: <https://www.flaticon.com/free-icons/camel>